

Colimits of Schur Rings

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A Bit of Notation

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Simple Quantities

If $C \subseteq G$, then

$$\overline{C} = \sum_{g \in C} g \in F[G].$$

Schur Rings

Definition - Schur Rings

Let $\mathcal{P}$ be a partition of finite support for the group G and let $\mathfrak{S} = \text{Span}\{\overline{C} \mid C \in \mathcal{P}\}$. We say $\mathfrak{S}$ is a **Schur Ring** (or S-ring) if

- (i) $\{1\} \in \mathcal{P}$,
- (ii) $C^* \in \mathcal{P}$ whenever $C \in \mathcal{P}$,
- (iii) $\overline{C} \overline{D} \in \mathfrak{S}$, whenever $C, D \in \mathcal{P}$.

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Schur Rings

The elements of a Schur partition $\mathcal{P}$ are called the **primitive sets** of $\mathfrak{S}$.

Any subset of G which is a union of primitive sets of S is called an **$\mathfrak{S}$ -subset**.

A subgroup $H \leq G$ which is also an S -set is called an **$\mathfrak{S}$ -subgroup**.

For an S -subgroup H , all the primitive sets contained in H form a Schur partition of H . The associated Schur ring over H is called a **Schur subring** of $\mathfrak{S}$.

Schur Rings

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Work by Bastian, Brewer, Humphries, M, and Thompson (2020) extended Schur rings to infinite groups.

Connections between Schur Rings and Association Schemes

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Association Schemes are “Group Theory without the Groups”

Connections between Schur Rings and Association Schemes

More generally, a Schur ring $\mathfrak{S} \subseteq \mathbb{Z}[G]$ coincides to the association scheme $\mathcal{A} = \left\{ \bigcup_{g \in C} R_g \mid C \in \mathcal{P} \right\}$, where $\mathcal{P}$ is the partition associated to S .

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The extension of Schur rings to infinite groups may prove to be the trailblazer for association schemes' path to the infinite.

Examples of Schur Rings

Examples of Schur Rings:

- i) Every group algebra $F[G]$ is a Schur ring with partition $G = \bigcup_{g \in G} \{g\}$, called the **discrete Schur ring**.

e.g. $G = \mathbb{Z}_6 = \langle a \rangle$:

$$S = \text{Span}\{ 1, a, a^2, a^3, a^4, a^5 \}$$

Examples of Schur Rings

Examples of Schur Rings:

- ii) Let $\mathcal{P} = \{\{1\}, G \setminus \{1\}\}$. This Schur ring is known as the **Trivial Schur ring**.

e.g. $G = \mathbb{Z}_6 = \langle a \rangle$:

$$S = \text{Span}\{ \textcolor{red}{1}, \textcolor{green}{a} + \textcolor{green}{a}^2 + \textcolor{green}{a}^3 + \textcolor{green}{a}^4 + \textcolor{green}{a}^5 \}$$

Examples of Schur Rings

Examples of Schur Rings:

- iii) Let $\mathcal{H} \leq \text{Aut}(G)$ and let S be the Schur ring afforded by the automorphism classes of $\mathcal{H}$, called an **automorphic Schur ring**.

e.g. $G = \mathbb{Z}_7 = \langle a \rangle$, $\mathcal{H} = \langle 2 \rangle$:

$$S = \text{Span}\{ \textcolor{red}{1}, \textcolor{green}{a} + \textcolor{green}{a}^2 + \textcolor{green}{a}^4, \textcolor{blue}{a}^3 + \textcolor{blue}{a}^5 + \textcolor{blue}{a}^6 \}$$

Examples of Schur Rings

Examples of Schur Rings:

- iv) Let $H \times K = G$. Let $\mathfrak{S}$ and $\mathfrak{T}$ be Schur rings over H and K , respectively. Then $\mathfrak{S} \otimes_F \mathfrak{T}$ is a Schur ring over G , called the **Direct Product** of $\mathfrak{S}$ and $\mathfrak{T}$. Its partition is constructed by all the possible products of sets from the partitions of $\mathfrak{S}$ and $\mathfrak{T}$.

e.g. $G = \mathbb{Z}_6 = \langle a \rangle$, $H = \mathbb{Z}_2$, $K = \mathbb{Z}_3$:

$$S = \text{Span}\{ \textcolor{red}{1}, \textcolor{green}{a^3}, \textcolor{blue}{a^2} + \textcolor{blue}{a^4}, \textcolor{violet}{a} + \textcolor{violet}{a^5} \}$$

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Examples of Schur Rings:

- v) Let $H \trianglelefteq G$. Let S and T be S-rings over H and G/H , respectively. The partition which affords T over G/H can be inflated into a partition of G . The combination of the partitions of S and T afford an S-ring, $S \wedge T$, over G , called the **Wedge Product** of S and T .

e.g. $G = \mathbb{Z}_6 = \langle a \rangle$, $H = \mathbb{Z}_2$, $K = \mathbb{Z}_3$:

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These Schur rings are called the **Traditional Schur rings**.

Traditional Schur Rings

Traditional Schur rings are the best known Schur rings and the easiest to construct, but many finite groups are known to have non-traditional Schur rings.

Theorem - Wielandt (1949), Shiu (1989)

There exists non-traditional Schur rings over dihedral groups.

Theorem - M (2014)

There exists non-traditional Schur rings over elementary abelian groups.

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Theorem - BBHMT (2020)

There exists non-traditional Schur rings over free groups, free products, and HNN extensions.

Theorem - Chen, He, Wu (2024)

There exists non-traditional Schur rings over free abelian group groups $\mathbb{Z} \times \mathbb{Z}$.

Traditional Schur Rings

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All Schur rings over finite cyclic groups are traditional.

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All Schur rings over the infinite cyclic group $\mathbb{Z}$ are traditional.

All Schur rings over the virtually cyclic group $\mathbb{Z} \times \mathbb{Z}_2$ are traditional.

Traditional Schur Rings

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A prime p is **Fermat** if $p = 2^k + 1$. There are probably only five such primes.

A prime p is **safe** if $p = 2q + 1$ and q is itself prime (a Sophie Germain prime). There are probably infinitely many safe primes.

Theorem - Bastian, M (2023)

All Schur rings over the virtually cyclic group $\mathbb{Z} \times \mathbb{Z}_p$ are traditional, where p is a Fermat or safe prime.

Traditional Schur Rings

Torsion-free, locally cyclic groups are subgroups of $\mathbb{Q}$.

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All Schur rings over the virtually locally cyclic group $G \times \mathbb{Z}_2$, where G is a torsion-free locally cyclic group, are traditional.

This proof relies on the compatibility of Schur subrings over infinite cyclic subgroups.

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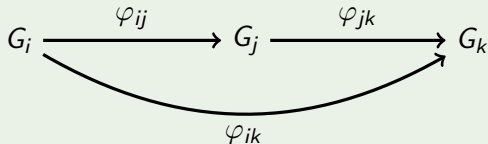
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Systems of Groups

Systems of Groups

Let $\mathcal{G} = \{G_i \mid i \in I\}$ be a set of groups G_i , indexed by a partially ordered set I . We say that $\mathcal{G}$ is a **system of groups** if there exists homomorphisms $\varphi_{ij} : G_i \rightarrow G_j$ such that:

- (1) φ_{ii} is the identity on G_i ;
- (2) $\varphi_{ik} = \varphi_{jk} \circ \varphi_{ij}$ whenever $i \leq j \leq k$, that is, the following diagram is commutative:

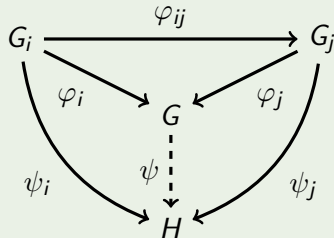


Colimits of Groups

The **colimit** of a system of groups $\mathcal{G}$, denoted $\lim_{\rightarrow} G_i$, is a group G equipped with homomorphisms $\varphi_i : G_i \rightarrow G$ such that:

- (1) $\varphi_i = \varphi_j \circ \varphi_{ij}$ whenever $i \leq j$;
- (2) if H is a **target** of $\mathcal{G}$, that is, a group with equipped homomorphisms $\psi_i : G_i \rightarrow H$ so that $\psi_i = \psi_j \circ \varphi_{ij}$, then there exists a unique homomorphism $\psi : G \rightarrow H$ so that $\psi_i = \psi \circ \varphi_i$.

In other words, the following diagram is commutative:



Examples of Colimits

- Unions of Chains
- Direct Limits (upper bounds exist)
- Coproduct (Free Products)
- Pushforwards (Free Products with Amalgamation)
- Coequalizers

Generally speaking, colimits are quotients of coproducts.

Locally cyclic groups are direct limits of cyclic groups.

Systems of Schur Rings

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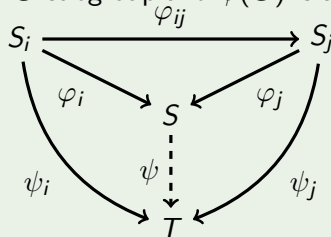
Let $\mathcal{G} = \{G_i \mid i \in I\}$ be a system of groups with maps $\varphi_{ij} : G_i \rightarrow G_j$. Let $\mathcal{S} = \{\mathfrak{S}_i \mid i \in I\}$ be a set of Schur rings $\mathfrak{S}_i$ over groups G_i . We say that $\mathcal{S}$ is a **system of Schur rings** over $\mathcal{G}$ if:

- (1) $\ker \varphi_{ij}$ is an $\mathfrak{S}_i$ -subgroup whenever $i \leq j$;
- (2) $\varphi_{ij}(\mathfrak{S}_i)$ is a Schur subring of S_j whenever $i \leq j$.

Colimits of Schur Rings

The **colimit** of a system of Schur rings $\mathcal{S}$, denoted $\varinjlim \mathfrak{S}_i$, is a Schur ring $\mathfrak{S}$ over $G = \varinjlim G_i$ such that:

- (1) $\ker \varphi_i$ is an $\mathfrak{S}_i$ -subgroup and $\varphi_i(\mathfrak{S}_i)$ is a Schur subring of $\mathfrak{S}$ for all i ;
- (2) if $\mathfrak{T}$ is a Schur ring over a target of $\mathcal{G}$, say H , and if $\ker \psi_i$ is an $\mathfrak{S}_i$ -subgroup and $\psi_i(\mathfrak{S}_i)$ is a Schur subring of $\mathfrak{T}$ for all i , then $\ker \psi$ is an $\mathfrak{S}$ -subgroup and $\psi(\mathfrak{S})$ is a Schur subring of $\mathfrak{T}$.



Cocompleteness

Theorem - M (2024)

Let $\mathcal{S} = \{\mathfrak{S}_i\}_i$ be a system of Schur rings over $\mathcal{G} = \{G_i\}_i$. Then the colimit $\lim_{\longrightarrow} S_i$ exists (and is unique) as a Schur ring over $\lim_{\longrightarrow} G_i$.

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The notion of colimits is defined for any category.

Cocompleteness

We say that a category is **cocomplete** if colimits exist for all diagrams.

The categories **Grp**, **Ab**, **Set**, **Vec**, **Top** are cocomplete.

Cocompleteness

Schur Homomorphisms

Let $\varphi : G \rightarrow H$ be a group homomorphism. Let $\mathfrak{S}$ be a Schur ring over G , and let $\mathfrak{T}$ be a Schur ring over H . We say φ is a **Schur homomorphism** if:

- (1) $\ker \varphi$ is an $\mathfrak{S}$ -subgroup;
- (2) $\varphi(\mathfrak{S})$ is a Schur subring of $\mathfrak{T}$.

Category of Schur Rings

We define a category **SRing** whose objects are Schur rings and whose morphisms are Schur homomorphisms.

The previous theorem restated say that **SRing** is cocomplete.

Every group is a direct limit of its finitely-generated subgroups, which is helpful for locally finite groups.

Similarly, every Schur rings is a direct limit of subgroups generated by its primitive sets.

The proof of traditional Schur rings over torsion-free, locally-cyclic groups is simplified using direct limits.

Theorem - M (2024)

If G is a torsion-free, locally cyclic group and all Schur rings over the virtually cyclic group $\mathbb{Z} \times \mathbb{Z}_n$ are traditional, then all Schur rings over $G \times \mathbb{Z}_n$ are traditional.

Direct limits of traditional Schur rings are (usually) traditional!

Torsion, locally cyclic groups are subgroups of $\mathbb{Q}/\mathbb{Z}$.

Theorem - M (2024)

All Schur rings over a locally cyclic group G are traditional.

Future Questions

What other virtually finite or locally finite groups might the Schur rings be classified over?

These techniques can be modified to the dual structure, limits.

Conjecture

All Schur rings over a pro-cyclic group, such as the p -adic integers, are traditional.

What other pro-finite groups might the Schur rings be classified over?

Thank you for listening!